



**ITM SKILLS
UNIVERSITY**

Department of Computer Science & Engineering

PROJECT REPORT

ShopifyFlow

Advanced Website Theme Editor

Student Name	Ashutosh Mishra
Roll Number	1500725129
Course / Branch	B.Tech - CSE
Academic Year	2025 – 2026

Index

#	Section	Page
1	Introduction & Overview	—
2	Student Details	—
3	Problem Statement	—
4	Technology Stack	—
5	Component Architecture	—
6	Must-Have Features	—
7	Screenshots & UI Walkthrough	—
8	Conclusion	—
9	References & Links	—

1. Introduction & Overview

ShopifyFlow is a React-based visual website editor inspired by Shopify's Theme Editor. It enables users to build and customize web pages using drag-and-drop functionality, real-time styling controls, smart design tools, and automatic CSS generation.

The project focuses on simplifying website creation by allowing users to visually arrange sections, customize styles, manage navigation menus, and instantly generate production-ready CSS without manually writing code.

ShopifyFlow demonstrates how a no-code visual editing experience can be built entirely on the front end using React's component model and Context API for shared state, making it a practical reference for website builders, landing page tools, and low-code platforms.

2. Student Details

Student Name	Ashutosh Mishra
Roll Number	1500725129
Course	B.Tech - CSE
Branch / Specialization	Computer Science & Engineering
College / University	ITM Skills University
Academic Year	2025 – 2026
Semester	VI
Subject	React js
Submission Date	June 2026

3. Problem Statement

Build a mock website editor (like Shopify's theme editor) where users can easily drag and arrange different parts of the website (like photo galleries or text boxes) and change their style using simple sliders.

The system should provide real-time editing capabilities, automatic CSS generation, dynamic menu management, template recovery mechanisms, and smart styling assistance — replicating the core experience of a professional theme editor in a lightweight, mock environment.

3.1 Objective

- **Build a fast, component-based drag-and-drop layout builder using React**
- **Provide real-time style controls (fonts, colors, spacing, layout) via a sidebar**
- **Keep navigation menus instantly in sync with page structure changes**
- **Generate clean, production-ready CSS automatically from visual edits**
- **Maintain a single shared design state across all editing tools**
- **Detect and auto-repair broken template sections for reliability**

3.2 Scope

This project covers the complete front-end implementation of ShopifyFlow, including the drag-and-drop canvas, style sidebar, menu builder, image picker, CSS generator, and template repair system. The current version operates as a client-side mock editor deployed on Vercel.

4. Technology Stack

Technology	Purpose	Version
React.js	Core UI library — component-based editor interface	18.2.0
React Context API	Centralized 'Current Design Center' shared state	Built-in
Vite	Development server and production build tool	4.4.9
JavaScript (ES6+)	Primary programming language	ES2022
HTML5	Semantic page structure	—
CSS3	Styling, layout, generated style output	—
Vercel	Hosting and CI/CD deployment	Cloud
Git & GitHub	Version control and source code management	—

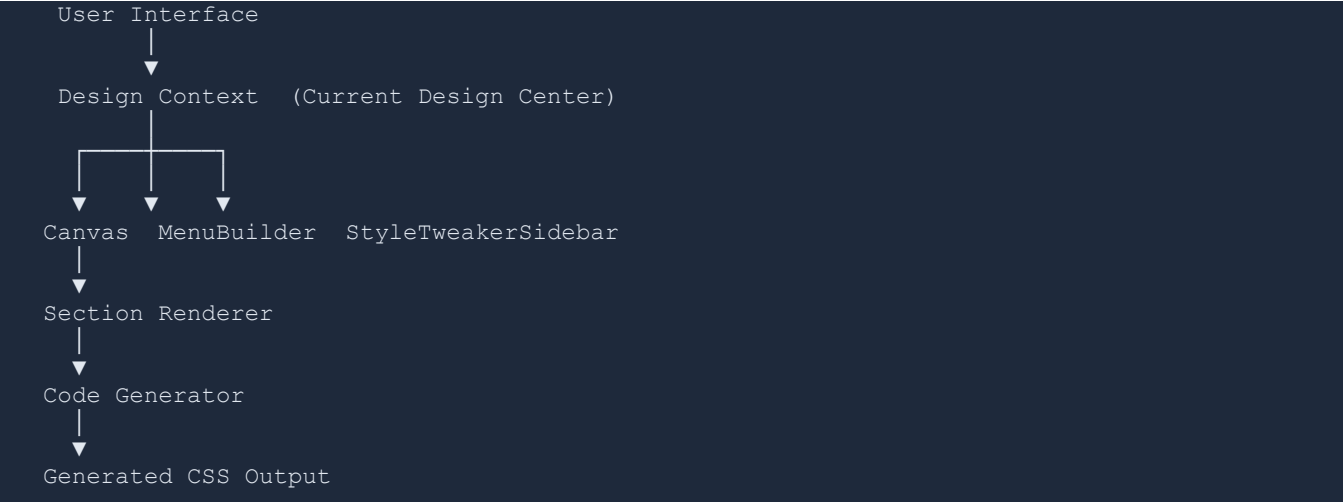
4.1 Why This Stack?

React 18 with Vite was chosen for fast HMR-driven development and a lightweight build pipeline well suited to an interactive editor UI. The React Context API powers the 'Current Design Center,' giving every tool — Canvas, MenuBuilder, and StyleTweakerSidebar — access to the same live design state without prop-drilling. Vercel provides zero-config, automatic deployments straight from GitHub.

5. Component Architecture

ShopifyFlow follows a centralized-state component architecture. A single Design Context shares the active template and its data with every editing tool, while a dedicated Section Renderer and Code Generator turn visual edits into live output.

5.1 Architecture Flow



5.2 Directory Structure

```
ShopifyFlow-Theme-Editor/
├── src/
│   ├── components/           # All editor UI components
│   ├── context/
│   │   └── DesignContext.jsx # Current Design Center
│   ├── hooks/
│   │   └── useDragDrop.js    # Drag-and-drop logic hook
│   ├── utils/
│   │   ├── colorUtils.js     # HEX validation helpers
│   │   └── generateCSS.js     # Smart CSS Code Generator logic
│   ├── data/
│   │   └── mockImages.js     # Smooth Image Picker dataset
│   ├── App.jsx               # Root component
│   ├── main.jsx              # SPA entry point
│   ├── index.html            # HTML entry
│   └── vite.config.js        # Vite configuration
```

5.3 Component Breakdown

Component	Responsibility	Type
App.jsx	Root component; mounts the editor and global providers	Root
DesignContext.jsx	Current Design Center — shares active template & design state	Context
Canvas.jsx	Main drag-and-drop work area where sections are arranged	Feature
DraggableSection.jsx	Wraps each section to make it draggable and droppable	UI

SectionRenderer.jsx	Renders the correct section type onto the canvas	Feature
StyleTweakerSidebar.jsx	Sidebar with sliders/pickers to adjust font, color, layout	Feature
MenuBuilder.jsx	Instant Menu Builder — add/remove/reorder nav links live	Feature
ImagePicker.jsx	Smooth Image Picker — browse and select from image library	Feature
CodeGenerator.jsx	Smart CSS Code Generator — converts styles into clean CSS	Feature
SmartColorInput.jsx	Validates HEX codes and shows a live color preview	UI
TemplateRepairKit.jsx	Detects and auto-restores broken template sections	Feature
TextEditor.jsx	Inline text editing for text-based sections	UI
Header.jsx / Footer.jsx	Top and bottom layout chrome for the editor shell	Layout
useDragDrop.js	Custom hook encapsulating drag-and-drop logic	Hook

6. Must-Have Features

ShopifyFlow implements the following core feature set, matching the original project brief:

Feature		Description
	Drag-and-Drop Layout	A screen area where you can grab different parts of the website (sections) and move them exactly where you want.
	Style Tweaker Sidebar	A panel on the side with controls (like sliders and color pickers) for changing the font size, color, or layout width of the selected element.
	Instant Menu Builder	A tool that updates the website's navigation menu instantly whenever you add or remove a page or link.
	Smooth Image Picker	A window that lets you quickly scroll through a massive library of product images without the browser slowing down.
	Code Generator	A smart feature that immediately translates your visual style choices into clean, ready-to-use CSS code.
	Current Design Center	A main system that shares the currently active website template and its basic structure with all the editing tools.
	Template Repair Kit	A safety system that automatically rebuilds and fixes a website section if a code error breaks its layout.
	Smart Color Input	A box that checks if you've typed a correct color code (like a Hex code) and shows you a preview of the color right away.

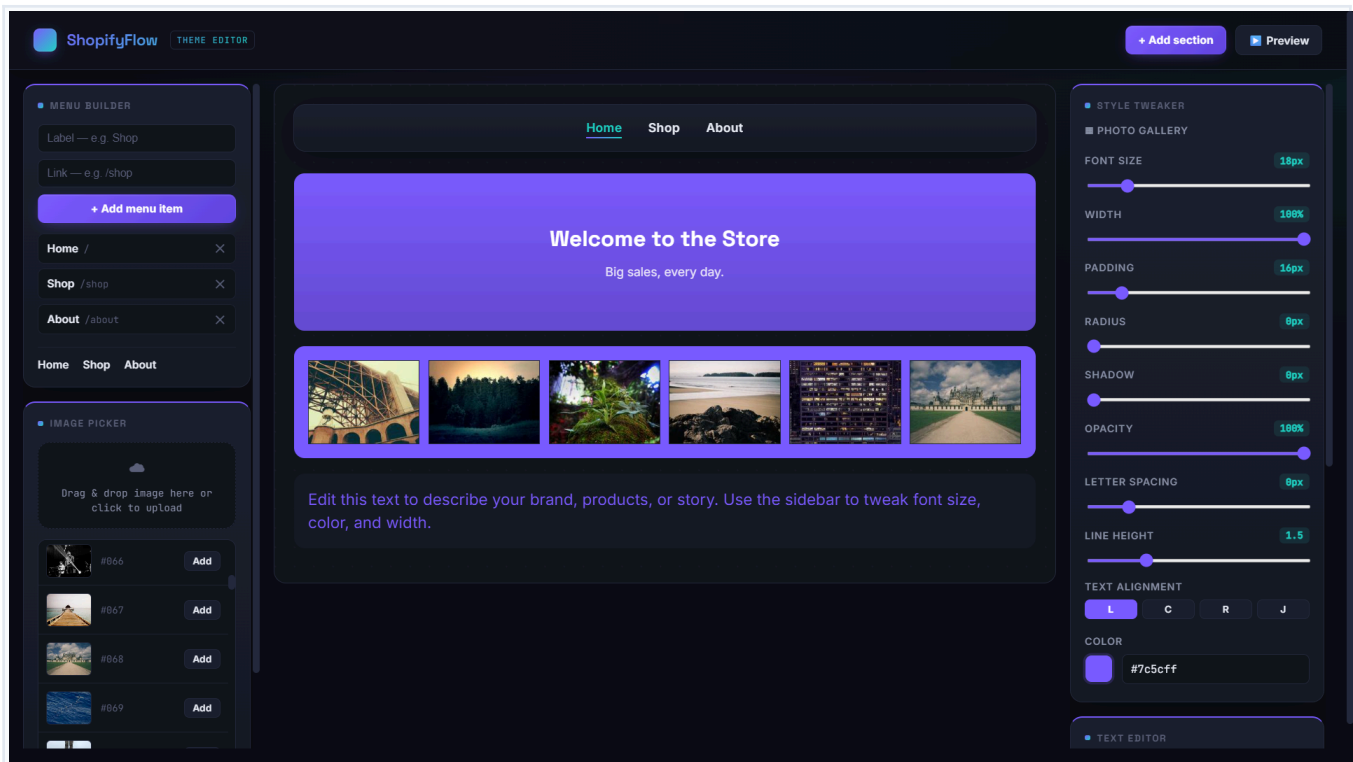
7. Screenshots & UI Walkthrough

The following section presents a walkthrough of the ShopifyFlow editor interface. As the platform is a live web application, screenshots can be captured from the deployed version at the URL below.

Live Platform: <https://shopifyflow-theme-editor.vercel.app/>

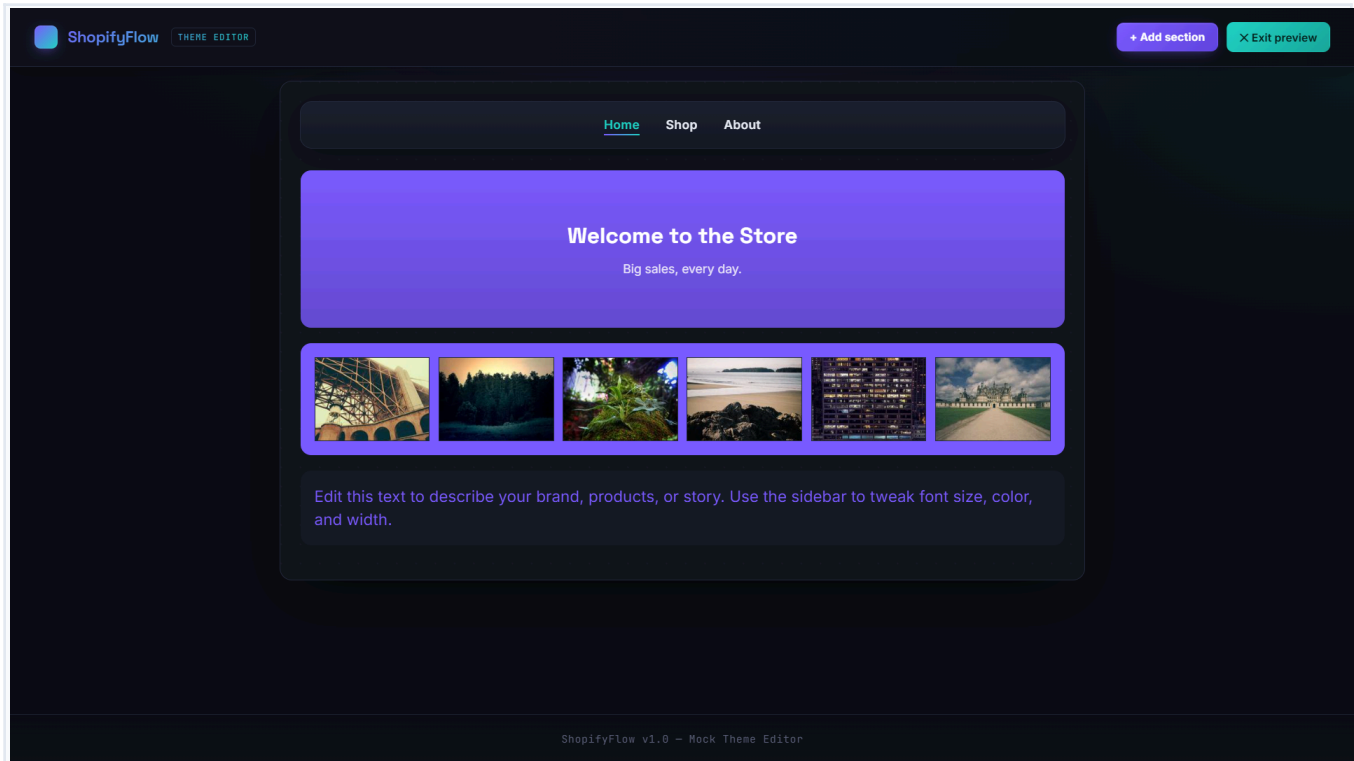
7.1 Editor Canvas (Drag-and-Drop)

The main editor view shows the Canvas in the center, where sections such as text blocks and image galleries can be dragged and dropped to rearrange the page layout in real time.



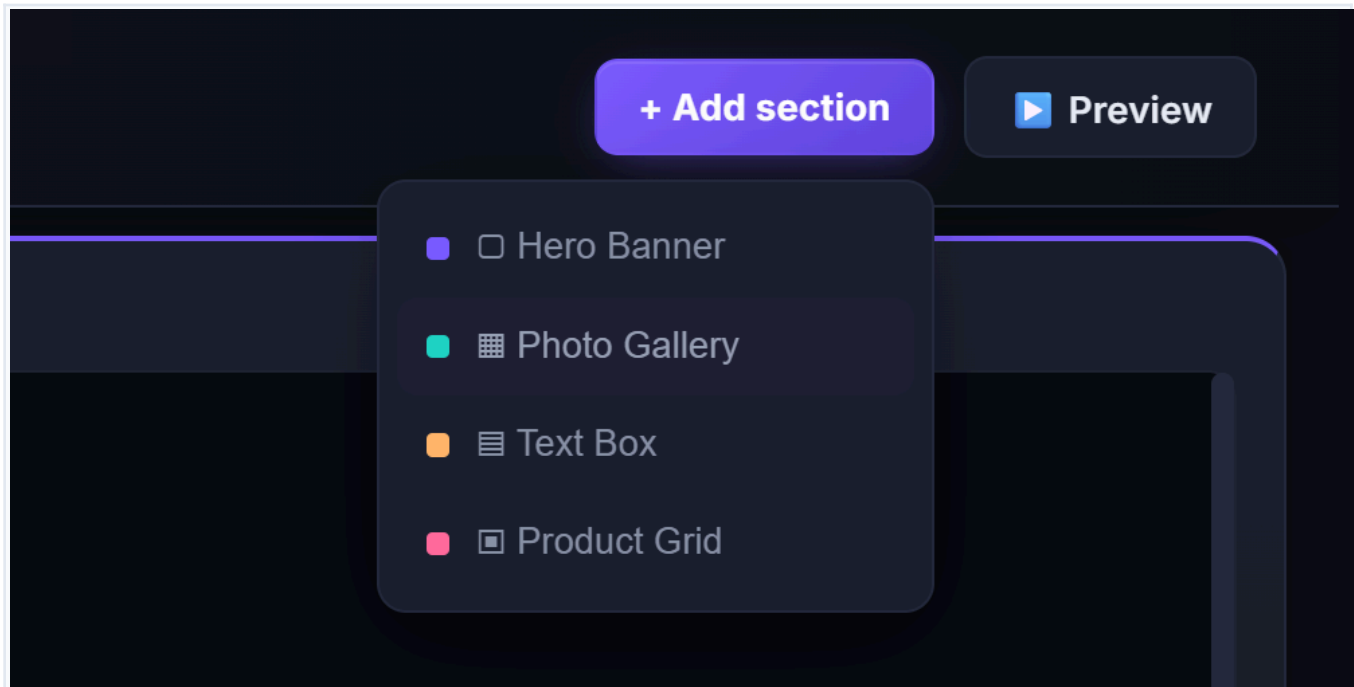
7.2 Web Preview

Here you can preview and check your website if they work properly or not..



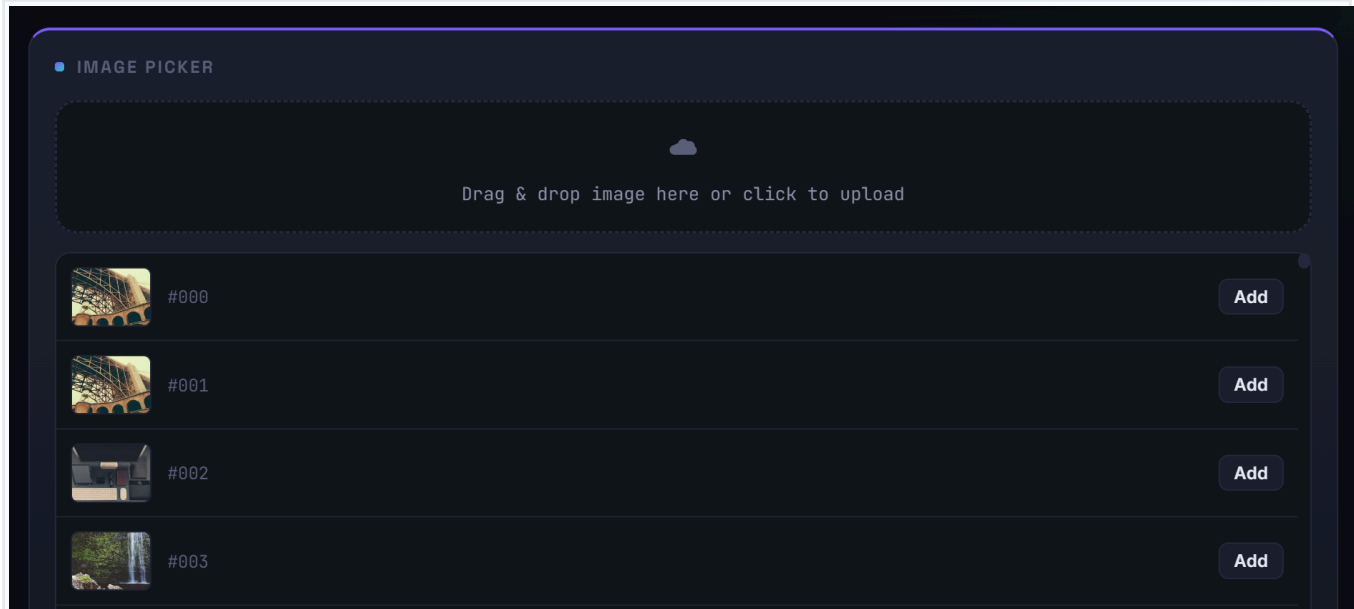
7.3 Instant Menu Builder

The Menu Builder panel lets users add, remove, and reorder navigation links, with the site's menu structure updating live as changes are made.



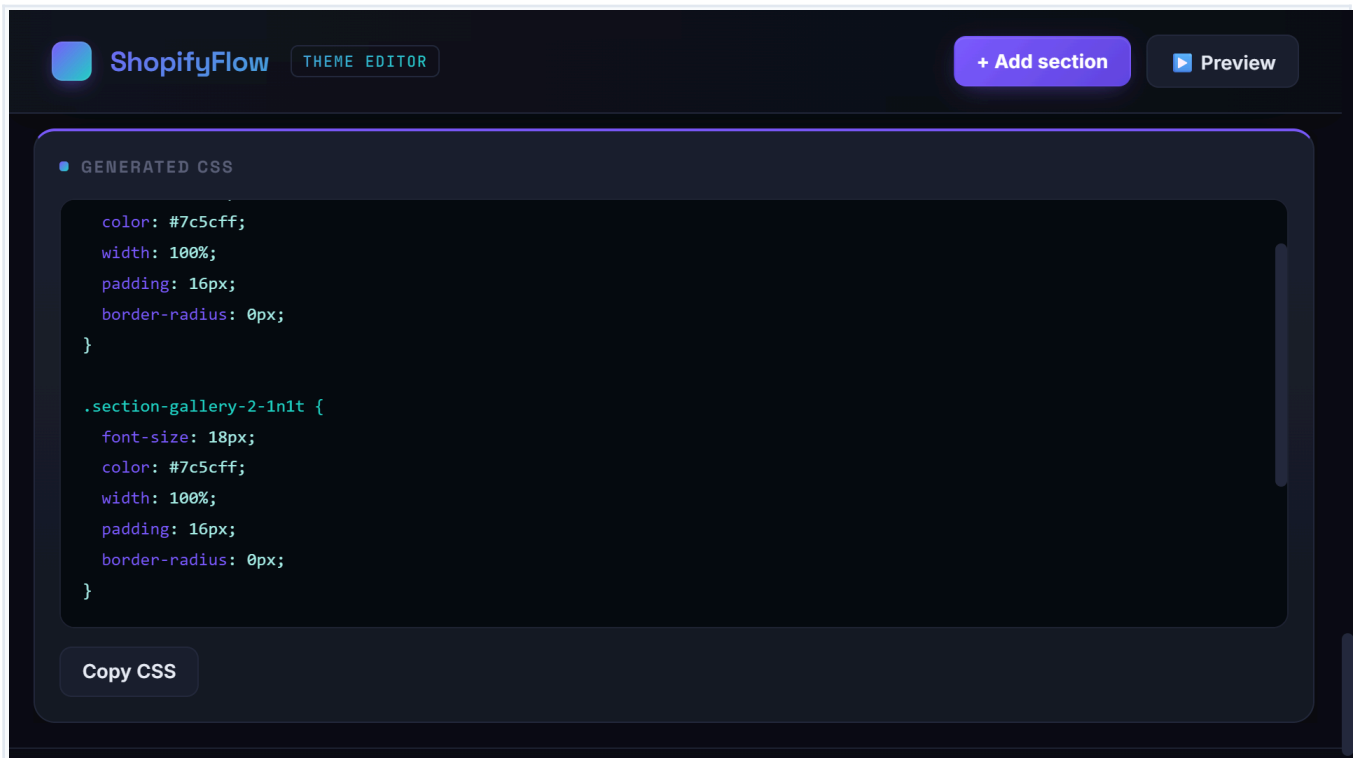
7.4 Smooth Image Picker

The Image Picker presents a scrollable library of mock product images, optimized so that browsing remains smooth even with a large image set.



7.5 Generated CSS Output

The Code Generator panel displays the clean, ready-to-use CSS that corresponds to the current visual design — bridging the visual editor with real code output.



8. Conclusion

ShopifyFlow successfully demonstrates a mock website theme editor that mirrors the core experience of professional tools like Shopify's Theme Editor, built entirely as a front-end React application. The project achieves all stated objectives:

- A working drag-and-drop layout builder for rearranging website sections
- A real-time Style Tweaker Sidebar for font, color, and layout adjustments
- An Instant Menu Builder that keeps navigation in sync with page structure
- A Smooth Image Picker capable of handling a large image library efficiently
- A Smart CSS Code Generator that turns visual edits into clean, usable CSS
- A centralized Current Design Center sharing state across all editing tools
- A Template Repair Kit that improves the editor's fault tolerance
- A Smart Color Input with HEX validation and live preview

The project reinforces key front-end engineering principles — centralized state management via Context API, component-based UI design, and translating visual interactions into generated code. It provides a strong foundation that could be extended with undo/redo, full HTML/CSS export, a theme marketplace, and user authentication for saved projects.

ShopifyFlow stands as a practical demonstration of how a no-code, visual website-building experience can be engineered using nothing but React, Context API, and Vite.

9. References & Links

9.1 Project Links

GitHub Repository	https://github.com/ashutosh062008/shopifyflow-theme-editor
Live Demo	https://shopifyflow-theme-editor.vercel.app/

9.2 Documentation References

- React Official Documentation — <https://react.dev>
- React Context API Documentation — <https://react.dev/learn/passing-data-deeply-with-context>
- Vite Build Tool Documentation — <https://vitejs.dev>
- MDN Web Docs — Drag and Drop API — <https://developer.mozilla.org>
- Vercel Deployment Documentation — <https://vercel.com/docs>

ShopifyFlow | Built with React + Vite | Deployed on Vercel | 2025-26